



Swinburne University of Technology Hawthorn Campus Department of Computing Technologies

COS20028 Big Data Architecture and Application Lab Task 6 - Semester 2, 2024

Assessment Title: Analysing App Download data with Pig.

Assessment Outcome: Pass/Fail (for practice)

Assessable Item:

- One (1) piece of a report containing the complete Pig Latin codes for preparing the data and finding the analytic result with the screenshot of submitted commands and execution results.

Purpose of Assignment

This assignment evaluates whether a student understands how to achieve an analytic goal via Apache Pig Latin with the local execution mode.

This is an individual assignment and is mandatory for passing this unit.

Introduction

The source data is built in the virtual machine image under the path of **/home/training/training_materials/analyst/data**. Our goal is using Apache Pig Latin to find out the answers to the given questions in the assignment task section in the local execution mode. You should only use local execution to complete the task.

The header of the input file with an example tuple is listed below:

App_ID	App_Name	App_Size	Price	Total_Rating	Avg_Rating	Category
281656475	PAC-MAN Premium	100788224	3.99	21292	4	Games

Assignment Goal

This assignment aims at training students to get familiar with Apache Pig Latin for managing data into the specific format and find required information in the local environment.

Assignment Task

The following tasks should be accomplished:

1. Load the **AppleStore.csv** data with Pig. Use only the "chararray," "int," and "double" data types to load the data. What are the first five (5) tuples in the loaded data with its default order and what is the bag structure of the result? Show the answers on the screen.
2. Find out only the free apps in the medical category. Converting your category values into lower case to prevent missing some tuples. Screenshot the last result after displaying the answer to the question on screen. You may encounter the system ask you to install fonts. Ignore it in the assignment.
3. How many Apps are in each category? Sort your result in the ascending order and display it on screen.

Please use the template to prepare your submission.

Submission Requirement

Only the submission using the given template will be evaluated by the tutors.

Failure to adhere to the submission requirements will immediately result in a failure for this assignment.

Rubric: Lab Task 6

Task	Assessment Outcome
Successfully finding the answer to the given question in the template and the submitted code in the report is reasonable, complete, and fully functional.	Pass/Fail

----- End of Assignment -----